

Document ID: TS-320

Title: Positioning Error Troubleshooting

Factory: FACTORY-DEMO-01

Station: S02

Classification: CONFIDENTIAL

Revision: 1.1

Effective Date: 2026-01-17

Owner: Demo Operations Engineering

Status: Approved

Synthetic demonstration document

Purpose and Scope

This procedure addresses POSITION-ENC-02 on station S02 Positioning.

Applies to P4711 investigations and repeated P4801/P4802/P4805/P4811 position warnings.

Observable symptoms

Positioning warning, failed homing reference, cycle interruption, downstream QUALITY-09 rejection.

Prerequisites and safety notes

Place S02 in maintenance mode. Follow local lockout and safety-zone procedures.

Numbered procedure

1. Record active alarms and product context.
2. Inspect encoder connector, cable routing, and reference mark.
3. Compare measured position against commanded reference.
4. Execute controlled homing and record result.

Encoder and positioning checks

Verify encoder feedback stability and mechanical coupling. Do not change restricted robot parameters.

Homing and dry cycle

Run homing after approved component replacement. Complete three dry cycles without POSITION-ENC-02.

Related events

P4711: S02 warning then S04 QUALITY-09 rejection. S02 alarm history includes E-STOP-17 escalation.

Verification and return-to-service criteria

Inspection verification passes, dry cycle is stable, and no active S02 position alarm remains.

Escalation criteria

Escalate if feedback remains unstable after encoder replacement or if repeated products fail the same position check.

Revision history

1.1 | 2026-01-17 | Added repeated S02 failure context | Demo Operations Engineering